DATA ANALYTICS PROJECT REPORT

Global Fuel Prices and Taxation Analysis

Interactive Tableau workbook and web dashboard exploring fuel prices, crude oil and recorded tax patterns across countries from January 2020 to April 2026.

Executive summary

This project examines petrol, diesel and LPG prices alongside Brent crude oil values and recorded fuel tax percentages. The provided dataset contains 27,468 observations for 84 countries across seven regions, with 327 distinct observation dates from 6 January 2020 to 6 April 2026. Tableau is used for the original analytical workbook; a responsive browser dashboard provides an additional way to filter and compare the same source data.

RECORDS	COUNTRIES	REGIONS	DATE POINTS
27,468	84	7	327
Source rows	Unique country values	Geographic groups	Distinct dates

Project objectives

- Compare fuel-price levels across countries and regions.
- Explore how observed fuel prices change over time and compare these movements with crude oil values.
- Summarize fuel-tax patterns at region and country level.
- Provide filters, charts, source files, SQL validation queries and project documentation for review.

Important scope note: the dataset ends on 6 April 2026; the project does not contain a complete calendar year of 2026.

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1. Dataset and data quality

The source is the CSV included in the repository and packaged Tableau workbook. The analysis uses the provided values as recorded. It has not been independently reconciled against official national or international statistics.

Column	Description
date	Observation date
country	Country name
region	Geographic region
income_level	Income classification in the supplied data
subsidy_level	Subsidy classification in the supplied data
petrol_usd_liter	Petrol price in USD per liter
diesel_usd_liter	Diesel price in USD per liter
lpg_usd_liter	LPG price in USD per liter
brent_crude_usd	Brent crude value in USD per barrel
tax_percentage	Recorded fuel tax percentage

Validation summary

Check	Result
Rows and columns	27,468 observations; 10 fields
Country and region coverage	84 countries; 7 regions
Observation dates	2020-01-06 to 2026-04-06
Distinct dates	327
Missing values	None found in the 10 columns inspected in the supplied CSV
Numeric ranges	Petrol 0.01-6.779 USD/L; diesel 0.01-6.240 USD/L; LPG 0.01-5.100 USD/L; Brent 73.735-130.000 USD/bbl; tax 0-65%

The observation dates are weekly, with 327 distinct dates across the available period. Aggregated monthly charts group all observations falling in each calendar month. Averages treat each source observation equally; they are not weighted by population, consumption or sales.

2. Descriptive findings

The values below are arithmetic means across all 27,468 source rows. They describe this dataset and should not be interpreted as current prices for a particular country or as official global estimates.

Metric	Mean in source dataset
Petrol price	USD 2.282 per liter
Diesel price	USD 2.134 per liter
LPG price	USD 1.711 per liter
Brent crude value	USD 107.982 per barrel
Recorded tax percentage	26.774%

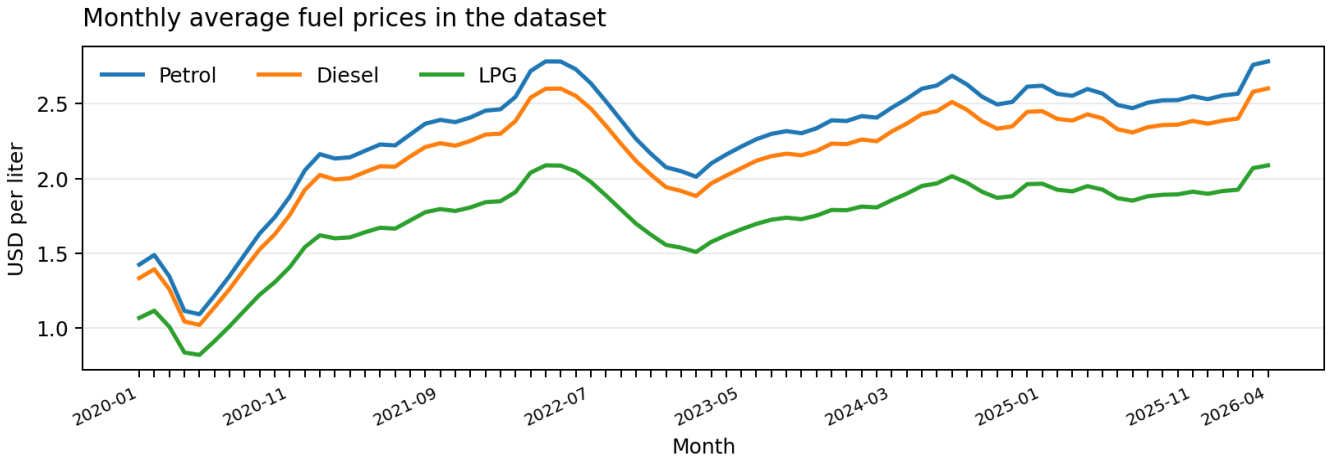


Figure 1. Monthly mean fuel prices across observations in each calendar month.

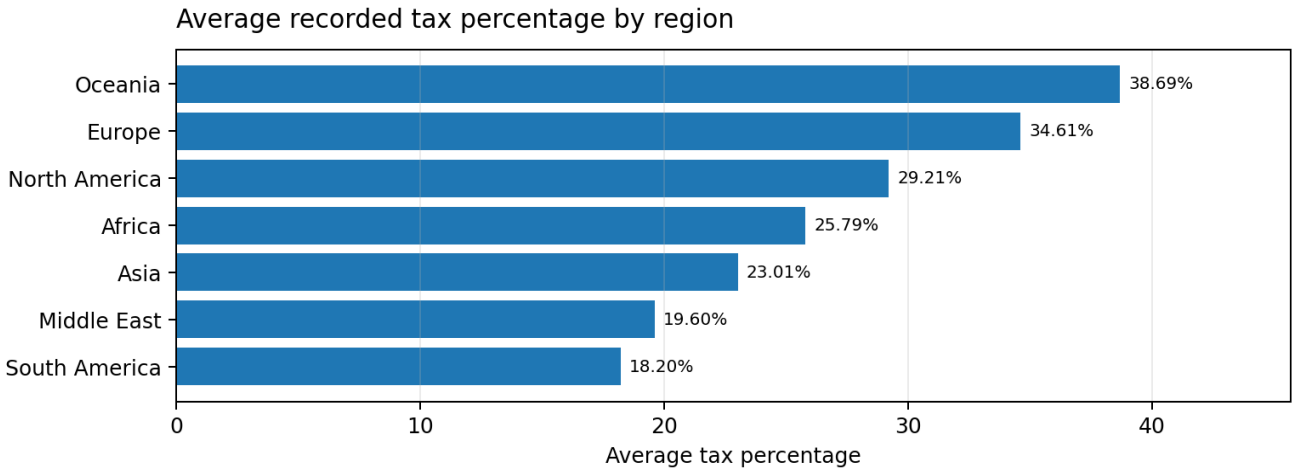


Figure 2. Average recorded tax percentage by region, calculated across all observations.

3. Country comparisons and deliverables

Hong Kong has the highest average petrol value in this dataset. Countries listed below are ranked by their mean across all available observations, not by a single current quote.

Country	Average petrol (USD/L)
Hong Kong	5.541
Norway	4.764
Netherlands	4.655
Ireland	4.654
Denmark	4.543
Singapore	4.543

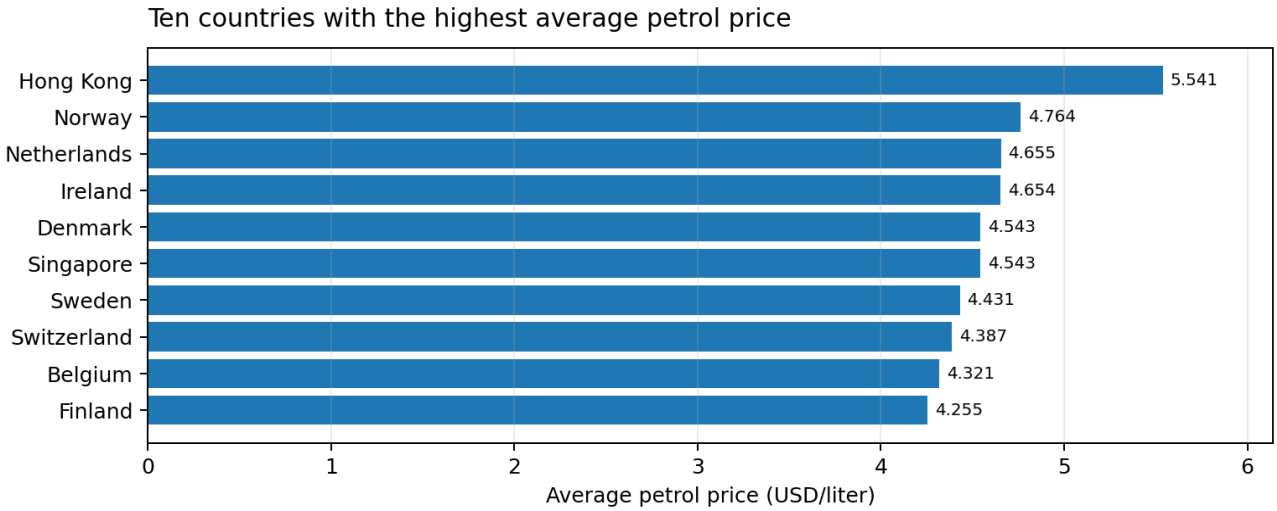


Figure 3. Top ten countries ranked by average petrol price in the supplied dataset.

Current Tableau workbook

The uploaded packaged workbook contains 14 distinct worksheet names, three regular dashboards and one Story with three saved points: Global Overview, Fuel Trends, and Tax & Country Analysis. The separate web dashboard is a native HTML/CSS/JavaScript implementation driven by the same CSV and is not an embedded Tableau Public view. The original Tableau workbook remains available for download.

4. Interactive web dashboard preview

The website dashboard is a responsive HTML/CSS/JavaScript implementation using the same CSV as the Tableau workbook. The screenshot below was captured from the browser rendering with the complete source dataset. It is a web-native dashboard, not a Tableau Public embed.

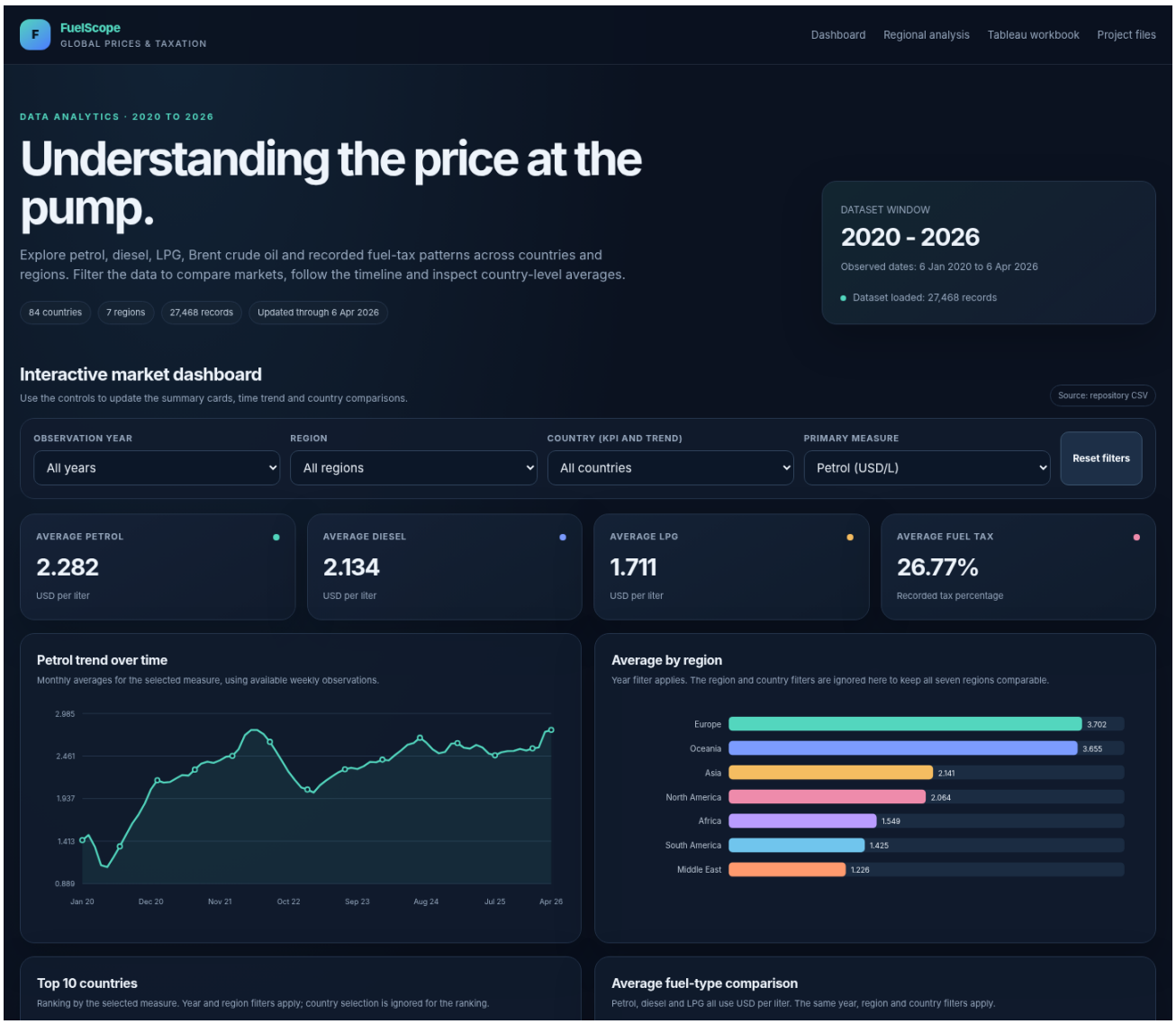


Figure 4. Browser dashboard preview; interactive filters are available in the hosted page.

The page provides year, region, country and primary-measure controls. KPI cards, a time trend, regional comparison, country ranking, fuel-type chart and regional summary table update in response to the applicable filters. The filtered rows can be exported as CSV.

Deployment note: the bundle is ready for GitHub Pages, but the site becomes public only after the repository files are committed and Pages is enabled by the repository owner.

5. Technical implementation and testing

Technology stack

• **MySQL 8.0:** local database import and validation workflow; example setup and validation queries are provided in the repository SQL folder.

• **Tableau:** packaged workbook with geographic maps, time-series views, KPI sheets, tax comparisons, three dashboards and a three-point Story.

• **Web dashboard:** responsive static HTML/CSS/JavaScript with custom SVG charts, filters for year/region/country/measure, and filtered CSV export. It reads the repository CSV at runtime.

• **GitHub Pages:** the website bundle is prepared for deployment from the repository root. It becomes publicly accessible only after the owner commits the files and enables Pages in repository settings.

Local browser smoke-test results

Check	Observed result
Test status	PASS
Rows loaded in browser	27,468
Dashboard ready time	1.187 seconds (local baseline)
Dashboard JS load metric	906 ms (local baseline)
Verified UI behavior	Four charts, 7-region table, year filter, region/country options and filtered CSV download
Runtime errors	No JavaScript runtime errors or browser console errors during the test

Test limitation: the browser test injects the local CSV through a test hook so it can run in a restricted environment without a local web server. Its timings exclude real HTTP transfer latency and are not a Lighthouse score, hosted-site benchmark or production service-level guarantee. Re-test the deployed GitHub Pages URL after publication.

6. Limitations and next steps

• The source data has not been independently verified against official statistics; use its provenance and licensing terms before redistribution.

• The dataset does not establish why a price or tax differs between countries; correlation is not evidence of causation.

• Public interactive web dashboard: upload the bundle and enable GitHub Pages. The current browser dashboard is data-driven; it does not claim to embed a Tableau Public view.

• For a full final submission, add a hosted-page performance run, final screenshots and demonstration-video link. Update the README if files or links change.

Repository: github.com/architchhabda10-cpu/global-fuel-prices-taxation-analysis